

Mid-Term Internship Report

Upgrading the Ume API at INRIA

Computer Science Degree, 3rd year

Academic Year 2025 – 2026

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1. Company Overview

Inria is the national research institute for digital sciences and technology. My internship is within the EVREF research team, which specializes in software evolution, engineering, and design. They place a particular focus on software quality and open-source tools.

2. Preparation and Anticipated Challenges

Before starting the internship, the main challenge was the programming language. The project relies entirely on Pharo, an object-oriented language and environment that I did not have the opportunity to learn at university. To avoid being completely lost, I learned the basics of the Pharo language on my own. Additionally, since the project is open-source, I spent some time reading the existing documentation to get an early idea of its architecture. This preparation allowed me to overcome my initial stress regarding the technical stack and ask the right questions from the very beginning.

3. Expectations

My main objective during this internship is to upgrade the current Ume API to improve its usability for developers. My primary expectations were to build confidence in my technical skills and learn to adapt quickly to a new technological ecosystem. I also wanted to learn how

to write cleaner and more maintainable code following industry standards. Finally, I aimed to develop my autonomy by implementing features from design to completion.

4. Planned Tasks

I am working on Ume, a Property-Based Testing (PBT) framework for Pharo. Property-Based Testing is an automatic and random test generator. Its main advantage is that it can generate a large number of tests to find interesting edge cases that could break the code.

I have handled multiple tasks over the past few weeks. First, I simplified the campaign definition of Ume to make the configuration more intuitive for users. Then, I added a progress bar running on a separate thread, which allows users to have real-time feedback on the campaign's progress without freezing the environment. I also implemented a shrinking method. Shrinking is the capacity to reduce a failing input to find the smallest possible case that causes the same bug. It is essential for users to understand the origin of a bug more easily so it can be debugged quickly.

5. Review

To sum up this first period, I am extremely satisfied with the trust and autonomy the team has given me. Starting an internship with a completely new programming language was intimidating at first. However, I managed to adapt quickly and no longer feel lost in the codebase. This experience has significantly boosted my self-confidence, proving to me that I am capable of learning fast and contributing effectively to a technical team. It strongly reinforces my desire to pursue my career path in software development.